EE-102: Electric Circuit Analysis

LECTURE NO 4:

- *Parallel DC Circuits*
- *Series-Parallel Circuits*

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Organization of Lecture

1. Introduction
2. Parallel Resistors
3. Parallel Circuits
4. Power distribution in a Parallel Circuit
5. Current Divider Rule (CDR)
6. Voltage Division in Parallel
7. Open and Short Circuits
8. Series-Parallel Circuits

1. Introduction

- Two network configurations, series and parallel, form the framework for some of the most complex network structures. In this Lecture we will examine the ***parallel circuit*** and all the methods and laws associated with this important configuration.

2. Parallel Resistors (1/16)

- ***The current is limited only by the resistor R. The higher the resistance, less is the current, and conversely, as determined by Ohm's law.***

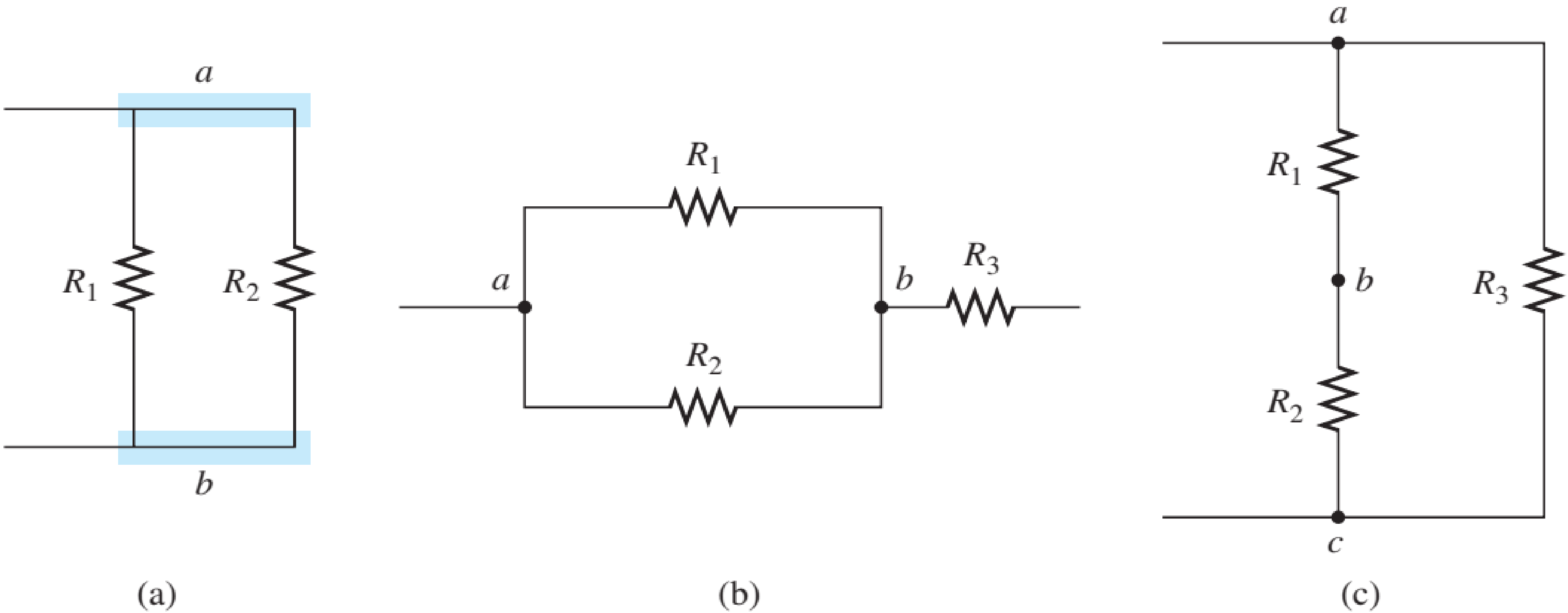

Figure 1:

(a) Parallel resistors; (b) R_1 and R_2 are in parallel; (c) R_3 is in parallel with the series combination of R_1 and R_2.

2. Parallel Resistors (2/16)

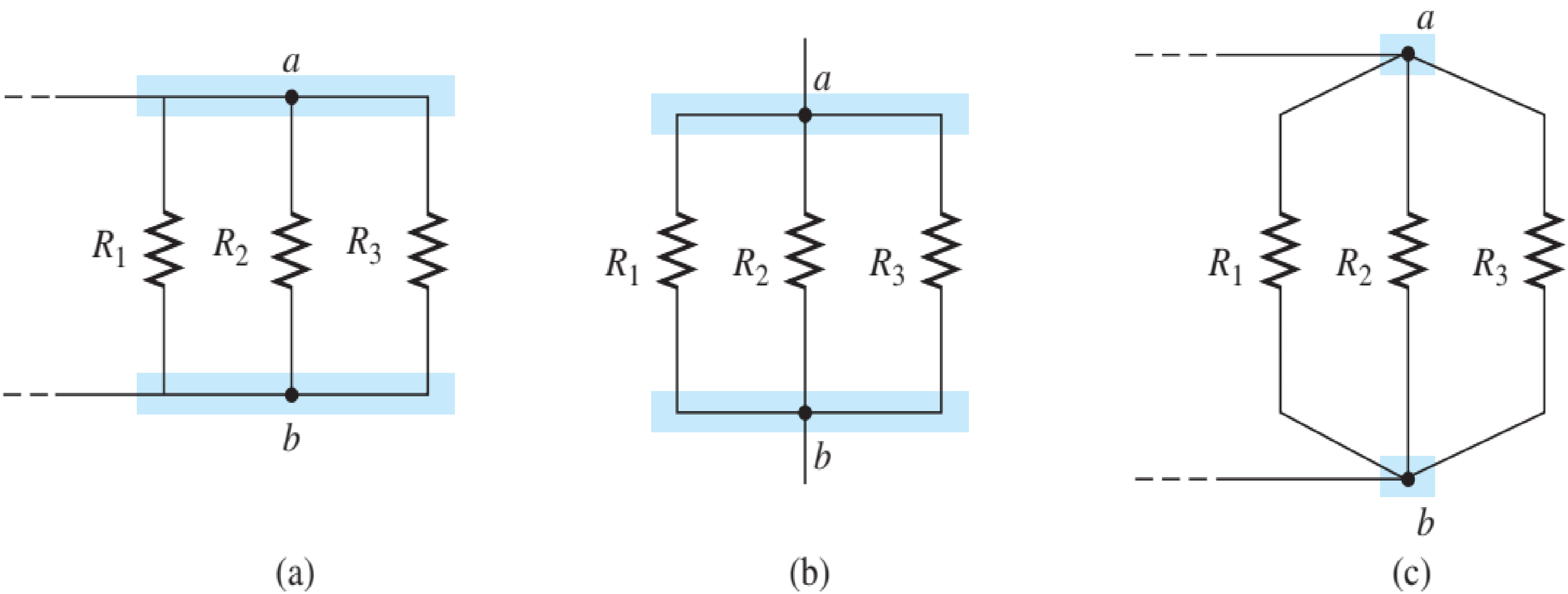

Figure 2

Schematic representations of three parallel resistors.

2. Parallel Resistors (3/16)

- For resistors in parallel as shown in Fig. 3,the total resistance is determined from the following equation:

$$\boxed{\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots + \frac{1}{R_N}} \qquad \textbf{(1)}$$

- Since G = 1/R, the equation can also be written in terms of conductance levels as follows:

$$\boxed{G_T = G_1 + G_2 + G_3 + \cdots + G_N} \qquad \text{(siemens, S)} \qquad \textbf{(2)}$$

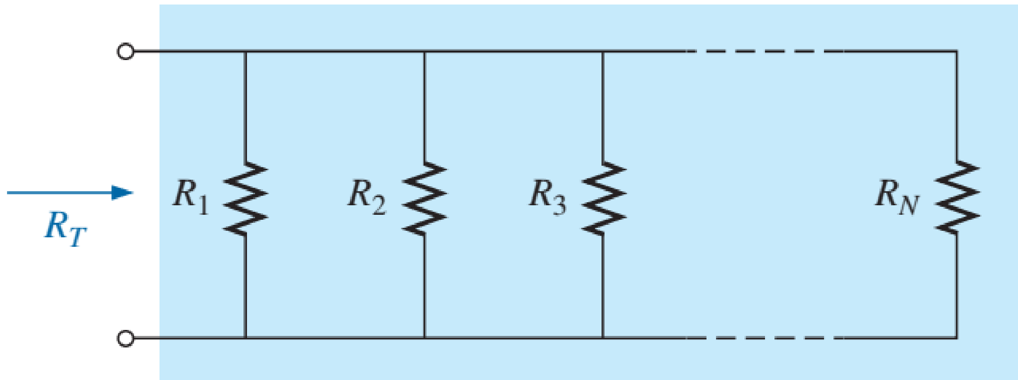

Figure 3

Parallel combination of resistors.

2. Parallel Resistors (4/16)

- In general, however, when the total resistance is desired, the following format is applied:

$$R_T = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots + \frac{1}{R_N}} \qquad \textbf{(3)}$$

2. Parallel Resistors (5/16)

EXAMPLE 1

a. Find the total conductance of the parallel network in Fig.4.

b. Find the total resistance of the same network using the results of part(a) and using equation 3.

Solutions:

a. $G_{1\pi} = \dfrac{1}{R_1} = \dfrac{1}{3\ \Omega} = 0.333\ \text{S}, \qquad G_2 = \dfrac{1}{R_2} = \dfrac{1}{6\ \Omega} = 0.167\ \text{S}$

and $G_T = G_1 + G_2 = 0.333\ \text{S} + 0.167\ \text{S} = \mathbf{0.5\ S}$

b. $R_T = \dfrac{1}{G_T} = \dfrac{1}{0.5\ \text{S}} = \mathbf{2\ \Omega}$

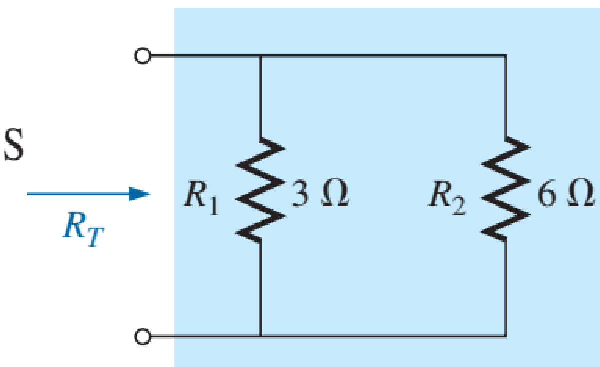

Figure 4
Parallel resistors for Example 1.

$$R_T = \frac{1}{\dfrac{1}{R_1} + \dfrac{1}{R_2}} = \frac{1}{\dfrac{1}{3\ \Omega} + \dfrac{1}{6\ \Omega}}$$

$$= \frac{1}{0.333\ \text{S} + 0.167\ \text{S}} = \frac{1}{0.5\ \text{S}} = \mathbf{2\ \Omega}$$

2. Parallel Resistors (6/16)

EXAMPLE 2

a. By inspection, which parallel element in Fig. 5 has the least conductance? Determine the total conductance of the network and note whether your conclusion was verified.

b. Determine the total resistance from the results of part (a) and by applying Eq. (3).

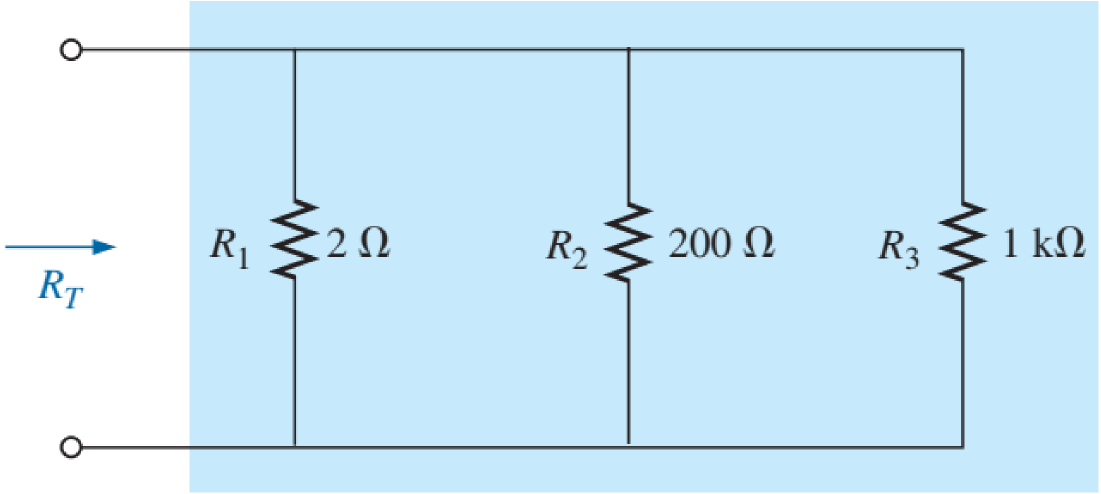

Figure 5

Parallel resistors

2. Parallel Resistors (6/16)

Solutions:

a. Since the 1 kΩ resistor has the largest resistance and therefore the largest opposition to the flow of charge (level of conductivity), it will have the lowest level of conductance:

$$G_1 = \frac{1}{R_1} = \frac{1}{2\ \Omega} = 0.5\ \text{S},\ G_2 = \frac{1}{R_2} + \frac{1}{200\ \Omega} = 0.005\ \text{S} = 5\ \text{mS}$$

$$G_3 = \frac{1}{R_3} = \frac{1}{1\ \text{k}\Omega} = \frac{1}{1000\ \Omega} = 0.001\ \text{S} = 1\ \text{mS}$$

$$G_T = G_1 + G_2 + G_3 = 0.5\ \text{S} + 5\ \text{mS} + 1\ \text{mS} = \mathbf{506\ mS}$$

Note the difference in conductance level between the 2 Ω (500 mS) and the 1 kΩ (1 mS) resistor.

b. $$R_T = \frac{1}{G_T} = \frac{1}{506\ \text{mS}} = \mathbf{1.976\ \Omega}$$

Applying Eq. (3) gives

$$R_T = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}} = \frac{1}{\frac{1}{2\ \Omega} + \frac{1}{200\ \Omega} + \frac{1}{1\ \text{k}\Omega}}$$

$$= \frac{1}{0.5\ \text{S} + 0.005\ \text{S} + 0.001\ \text{S}} = \frac{1}{0.506\ \text{S}} = \mathbf{1.98\ \Omega}$$

2. Parallel Resistors (7/16)

Homework!

EXAMPLE 3 Find the total resistance of the configuration in Fig. 6.

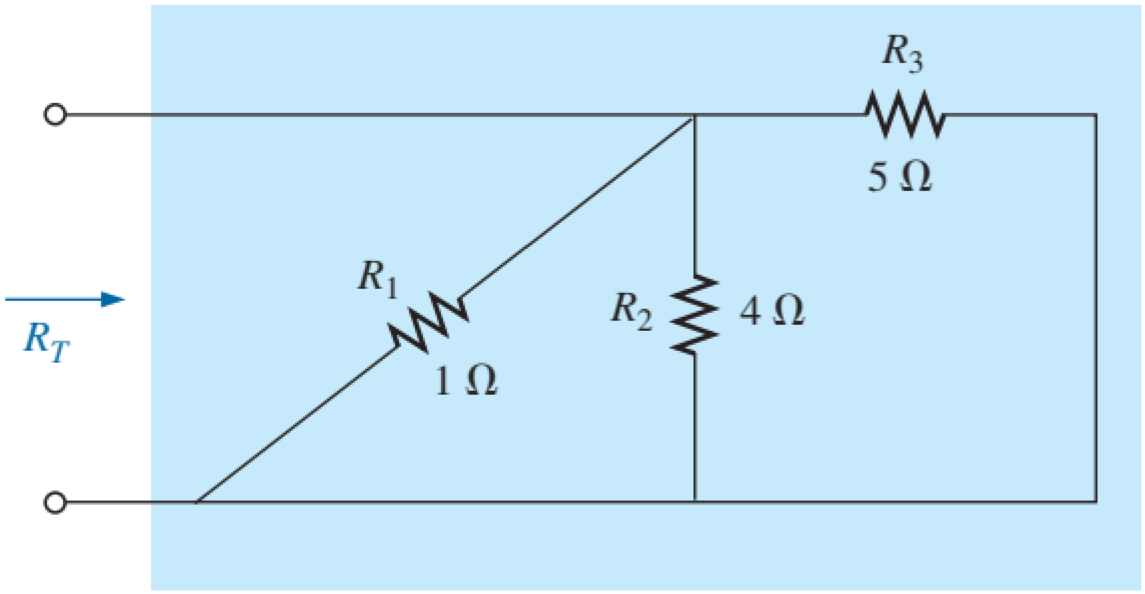

FIG. 6
Network to be investigated in Example 3.

2. Parallel Resistors (8/16)

EXAMPLE 4

a. What is the effect of adding another resistor of 100 Ω in parallel with the parallel resistors of Example 1 as shown in Fig. 7?

b. What is the effect of adding a parallel 1 Ω resistor to the configuration in Fig. 1?

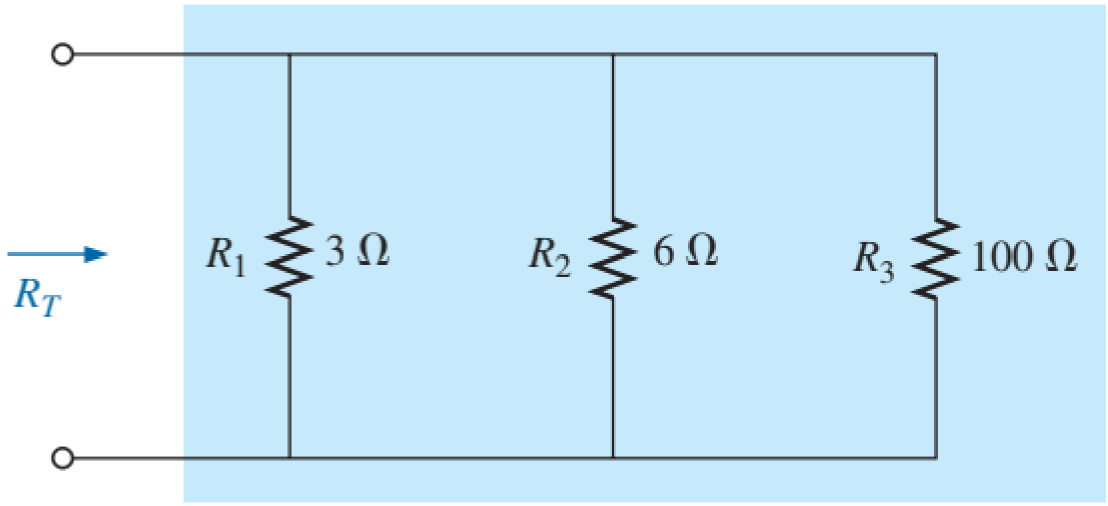

Figure 7

Adding a parallel 100 Ω resistor to the network in Fig. 4.

2. Parallel Resistors (9/16)

Solutions:

a. Applying Eq. (3) gives

$$R_T = \frac{1}{\dfrac{1}{R_1} + \dfrac{1}{R_2} + \dfrac{1}{R_3}} = \frac{1}{\dfrac{1}{3\ \Omega} + \dfrac{1}{6\ \Omega} + \dfrac{1}{100\ \Omega}}$$

$$= \frac{1}{0.333\ \text{S} + 0.167\ \text{S} + 0.010\ \text{S}} = \frac{1}{0.510\ \text{S}} = \mathbf{1.96\ \Omega}$$

- The parallel combination of the 3 Ω and 6 Ω resistors resulted in a total resistance of 2 Ω in Example 1. The effect of adding a resistor in parallel of 100 Ω had little effect on the total resistance. The total change in resistance was less than 2%. However, note that the total resistance dropped with the addition of the 100 Ω resistor.

2. Parallel Resistors (10/16)

b. Applying Eq. (3) gives

$$R_T = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4}} = \frac{1}{\frac{1}{3\ \Omega} + \frac{1}{6\ \Omega} + \frac{1}{100\ \Omega} + \frac{1}{1\ \Omega}}$$

$$= \frac{1}{0.333\ \text{S} + 0.167\ \text{S} + 0.010\ \text{S} + 1\ \text{S}} = \frac{1}{0.51\ \text{S}} = \mathbf{0.66\ \Omega}$$

- The introduction of the 1 Ω resistor reduced the total resistance from 2 Ω to only 0.66 — a decrease of almost 67%.
  - ***The total resistance of parallel resistors will always drop as new resistors are added in parallel, irrespective of their value.***
- For equal resistors in parallel, the equation for the total resistance becomes significantly easier to apply. For ***N*** equal resistors in parallel, Eq. (3) becomes:

$$\boxed{R_T = \frac{R}{N}} \qquad \mathbf{(4)}$$

2. Parallel Resistors (11/16)

EXAMPLE 5 Find the total resistance of the parallel resistors in Fig. 9.

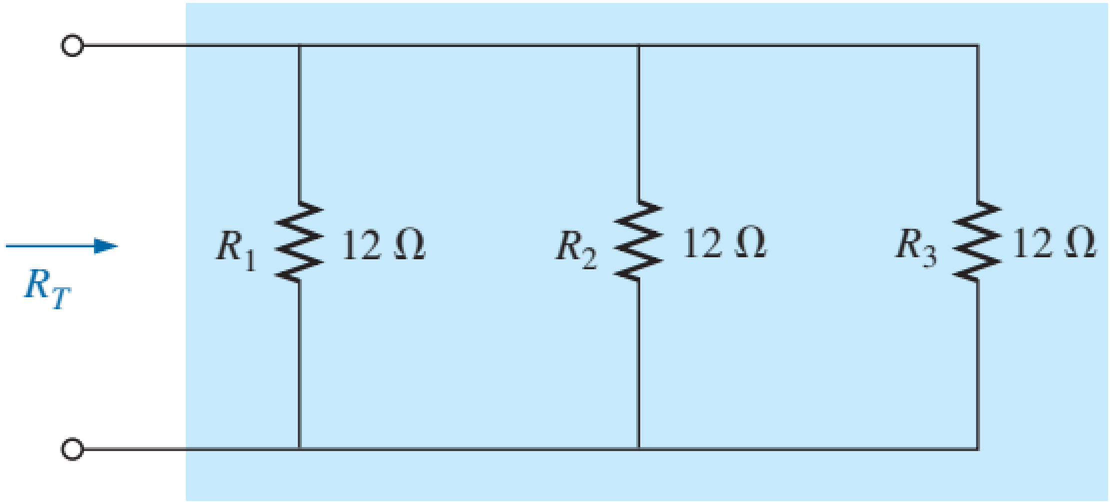

Fig. 9.

Solution: Applying Eq. (4) gives

$$R_T = \frac{R}{N} = \frac{12\ \Omega}{3} = \mathbf{4\ \Omega}$$

2. Parallel Resistors (12/16)

Homework!

EXAMPLE 6 Find the total resistance for the configuration in Fig. 10.

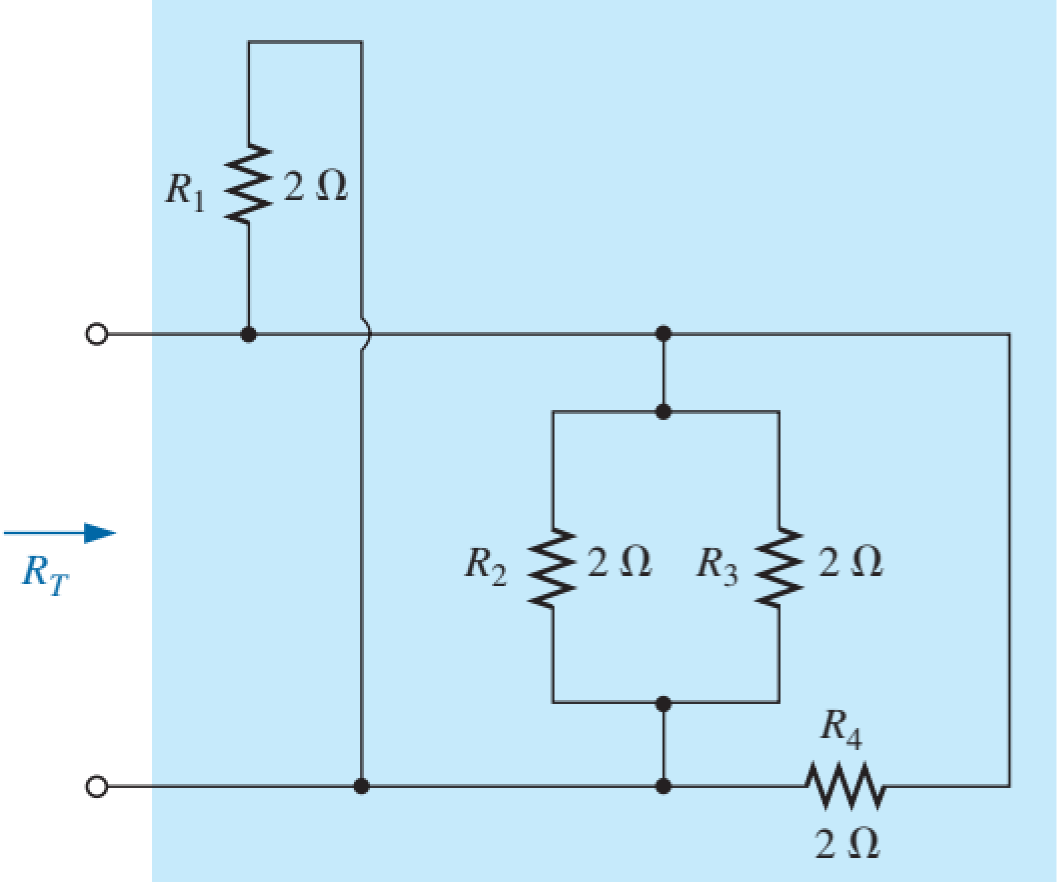

Fig. 10.

2. Parallel Resistors (13/16)

- Recall that series elements can be interchanged without affecting the magnitude of the total resistance. In parallel networks:
  - ***Parallel resistors can be interchanged without affecting the total resistance.***

2. Parallel Resistors (14/16)

Example 7: Determine the total resistance of the parallel elements in Fig. 11.

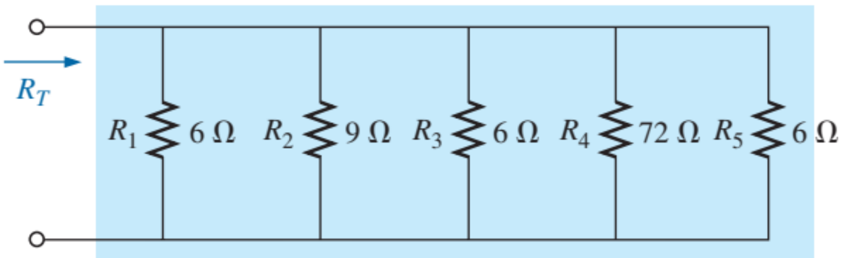

Figure 11

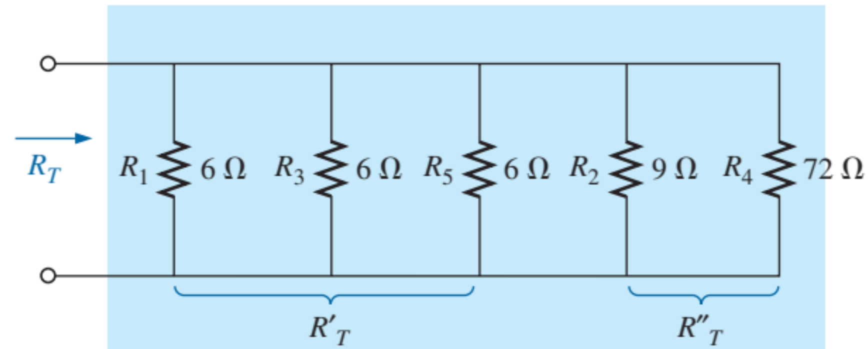

Figure 12

Solution:

$$R'_T = \frac{R}{N} = \frac{6\ \Omega}{3} = 2\ \Omega$$

$$R''_T = \frac{R_2R_4}{R_2 + R_4} = \frac{(9\ \Omega)(72\ \Omega)}{9\ \Omega + 72\ \Omega} = \frac{648}{81}\ \Omega = 8\ \Omega$$

$$R_T = \frac{R'_T R''_T}{R'_T + R''_T} = \frac{(2\ \Omega)(8\ \Omega)}{2\ \Omega + 8\ \Omega} = \frac{16}{10}\ \Omega = \mathbf{1.6\ \Omega}$$

2. Parallel Resistors (15/16)

Example 8: Determine the value of R_2 in Fig. 13 to establish a total resistance of 9 kΩ.

Solution:

$$R_T = \frac{R_1 R_2}{R_1 + R_2}$$

$$R_T(R_1 + R_2) = R_1 R_2$$

$$R_T R_1 + R_T R_2 = R_1 R_2$$

$$R_T R_1 = R_1 R_2 - R_T R_2$$

$$R_T R_1 = (R_1 - R_T) R_2$$

and

$$R_2 = \frac{R_T R_1}{R_1 - R_T}$$

Substituting values gives

$$R_2 = \frac{(9\ \text{k}\Omega)(12\ \text{k}\Omega)}{12\ \text{k}\Omega - 9\ \text{k}\Omega} = \frac{108}{3}\ \text{k}\Omega = \mathbf{36\ k\Omega}$$

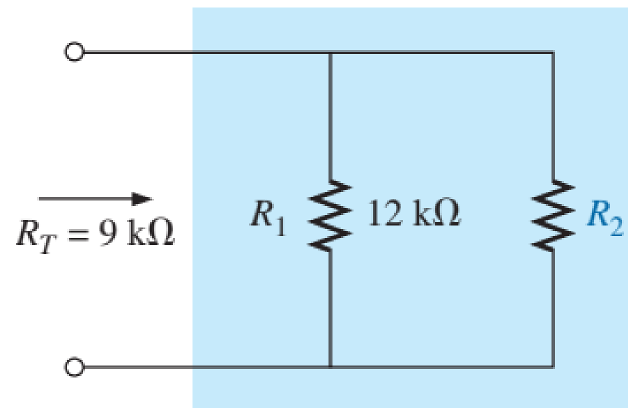

Figure 13

2. Parallel Resistors (16/16)

Homework!

Example 9: Determine the values of R_1 , R_2 , and R_3 in Fig. 14, if $R_2 = 2R_1$, $R_3 = 2R_2$, And the total resistance is 16 kΩ.

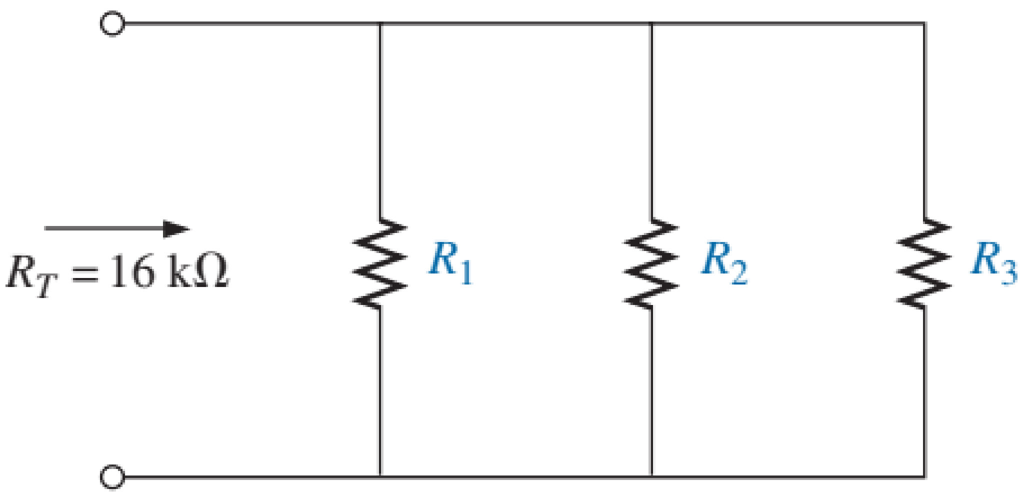

Figure 14

3. Parallel Circuits (1/7)

- A *parallel circuit* can now be established by connecting a supply across a set of parallel resistors as shown in Fig. 15. The positive terminal of the supply is directly connected to the top of each resistor, while the negative terminal is connected to the bottom of each resistor. Therefore, it should be quite clear that the applied voltage is the same across each resistor. In general:
  - ***The voltage is always the same across parallel elements.***
- For the voltages of the circuit in Fig. 18, the result is that:

$$\boxed{V_1 = V_2 = E} \qquad \textbf{(5)}$$

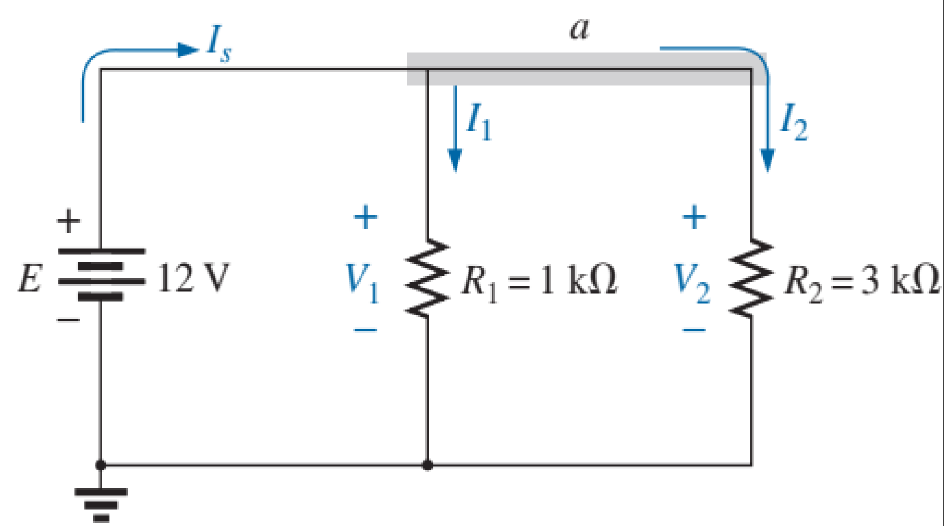

Figure 15

3. Parallel Circuits (2/7)

- Recall from series circuits that the source does not "see" the parallel combination of elements. It reacts only to the total resistance of the circuit, as shown in Fig. 16. The source current can then be determined using Ohm's law:

$$\boxed{I_s = \frac{E}{R_T}} \quad \textbf{(6)}$$

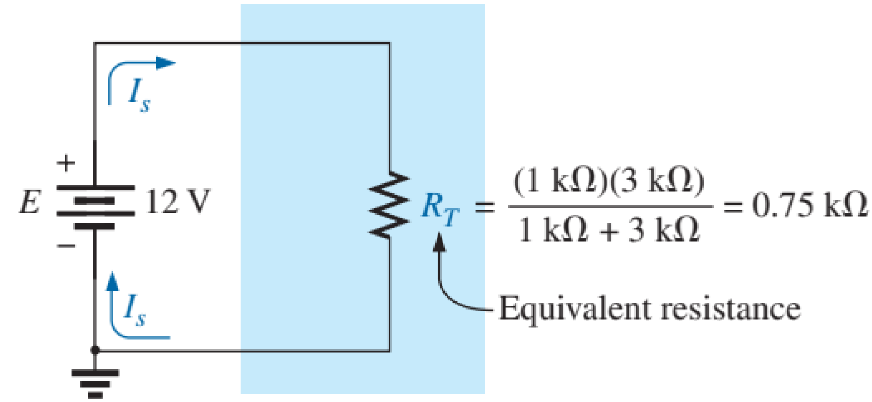

Figure 16

- Since the voltage is the same across parallel elements, the current through each resistor can also be determined using Ohm's law. That is,

$$\boxed{I_1 = \frac{V_1}{R_1} = \frac{E}{R_1} \quad \text{and} \quad I_2 = \frac{V_2}{R_2} = \frac{E}{R_2}} \quad \textbf{(7)}$$

3. Parallel Circuits (3/7)

- The relationship between the source current and the parallel resistor currents can be derived by simply taking the equation for the total resistance in Eq. (1):

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$$

Multiplying both sides by the applied voltage give

$$E\left(\frac{1}{R_T}\right) = E\left(\frac{1}{R_1} + \frac{1}{R_2}\right)$$

resulting in

$$\frac{E}{R_T} = \frac{E}{R_1} + \frac{E}{R_2}$$

Then note that $E/R_1 = I_1$ and $E/R_2 = I_2$ to obtain

$$\boxed{I_s = I_1 + I_2} \qquad \textbf{(8)}$$

The result reveals a very important property of parallel circuits:

For single-source parallel networks, the source current (I_s) is always equal to the sum of the individual branch currents.

3. Parallel Circuits (4/7)

Example 10: For the parallel network in Fig. 17:

a. Find the total resistance.

b. Calculate the source current.

c. Determine the current through each parallel branch.

d. Show that Eq. (8) is satisfied.

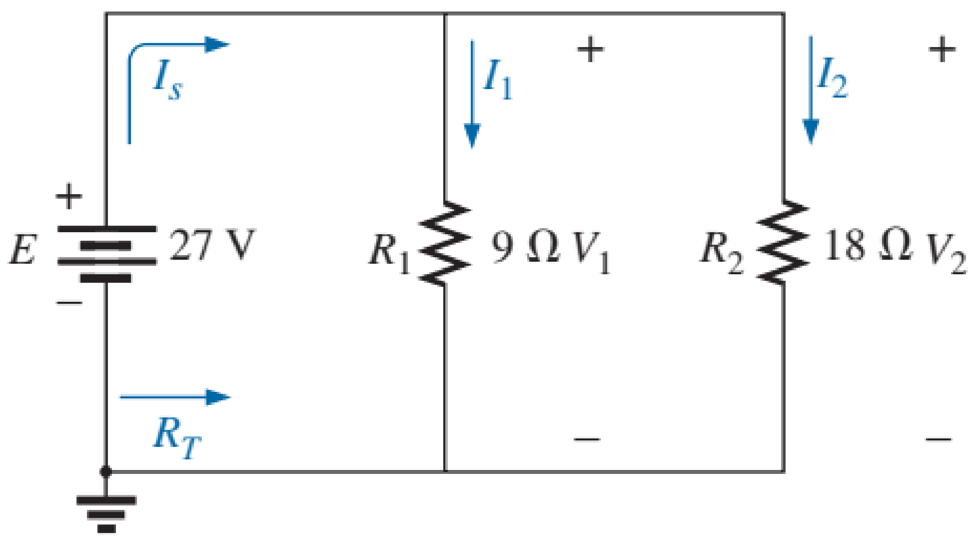

Figure 17

3. Parallel Circuits (5/7)

Solutions:

a. $$R_T = \frac{R_1 R_2}{R_1 + R_2} = \frac{(9\ \Omega)(18\ \Omega)}{9\ \Omega + 18\ \Omega} = \frac{162}{27}\ \Omega = \mathbf{6\ \Omega}$$

b. Applying Ohm's law gives

$$I_s = \frac{E}{R_T} = \frac{27\ \text{V}}{6\ \Omega} = \mathbf{4.5\ A}$$

c. Applying Ohm's law gives

$$I_1 = \frac{V_1}{R_1} = \frac{E}{R_1} = \frac{27\ \text{V}}{9\ \Omega} = \mathbf{3\ A}$$

$$I_2 = \frac{V_2}{R_2} = \frac{E}{R_2} = \frac{27\ \text{V}}{18\ \Omega} = \mathbf{1.5\ A}$$

d. Substituting values from parts (b) and (c) gives

$$I_s = \mathbf{4.5\ A} = I_1 + I_2 = 3\ \text{A} + 1.5\ \text{A} = \mathbf{4.5\ A} \qquad \text{(checks)}$$

3. Parallel Circuits (6/7)

Homework!

Example 11: For the parallel network in Fig. 18:

a. Find the total resistance.

b. Calculate the source current.

c. Determine the current through each branch.

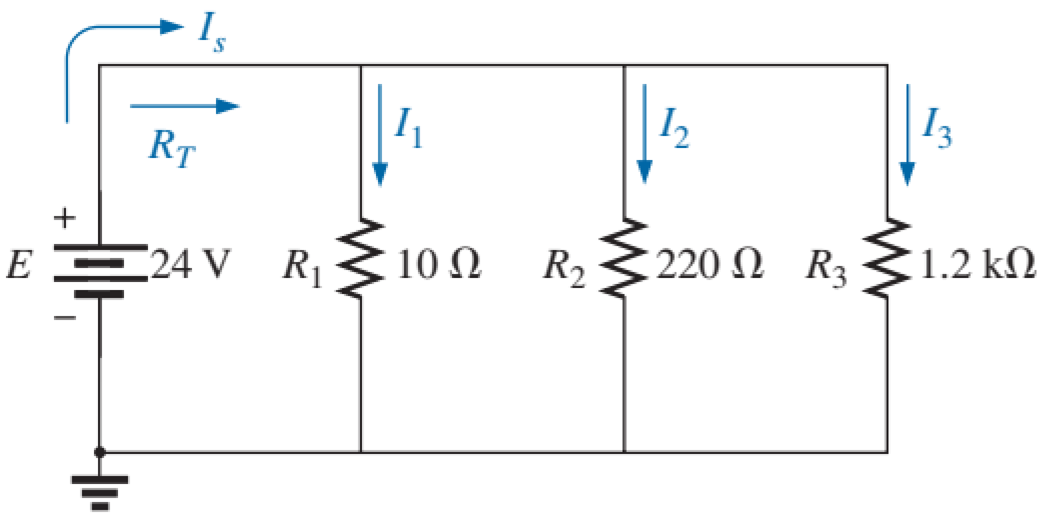

Figure 18

3. Parallel Circuits (7/7)

Homework!

Example 12: For the parallel network in Fig. 19:

a. Determine R_3 .

b. Find the applied voltage E.

c. Find the source current I_s.

d. Find I_2.

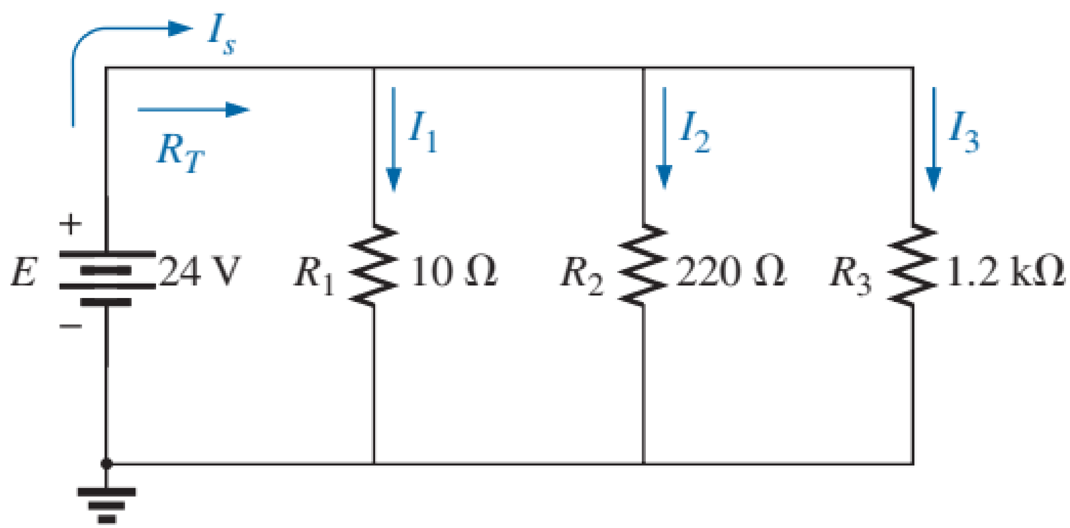

Figure 18

4. Power Distribution in a Parallel Circuit (1/9)

- ***For any network composed of resistive elements, the power applied by the battery will equal that dissipated by the resistive elements.***
- For the parallel circuit in Fig. 19:

$$\boxed{P_E = P_{R_1} + P_{R_2} + P_{R_3}} \qquad \textbf{(9)}$$

which is exactly the same as obtained for the series combination.

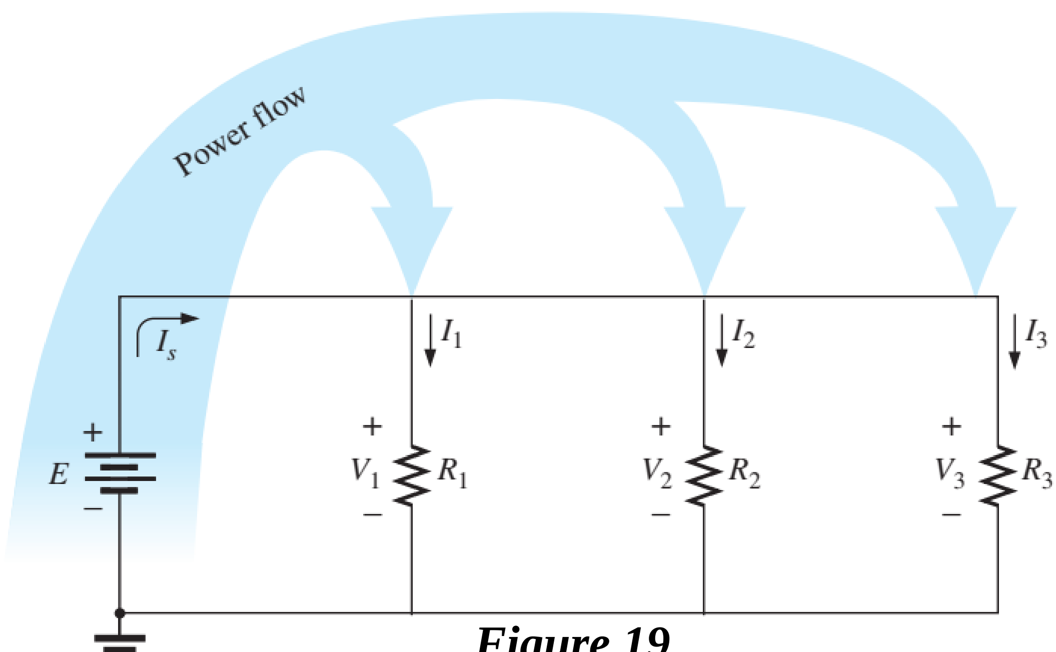

Figure 19

4. Power Distribution in a Parallel Circuit (2/9)

The power delivered by the source in the same:

$$\boxed{P_E = EI_s} \qquad \text{(watts, W)} \qquad \textbf{(10)}$$

as is the equation for the power to each resistor (shown for R_1 only):

$$\boxed{P_1 = V_1 I_1 = I_1^2 R_1 = \frac{V_1^2}{R_1}} \qquad \text{(watts, W)} \qquad \textbf{(11)}$$

In the equation $P = V^2/R$, the voltage across each resistor in a parallel circuit will be the same. The only factor that changes is the resistance in the denominator of the equation. The result is that

in a parallel resistive network, the larger the resistor, the less is the power absorbed.

4. Power Distribution in a Parallel Circuit (3/9)

Example 13: For the parallel network in Fig. 20:

a. Determine the total resistance R_T.
b. Find the source current and the current through each resistor.
c. Calculate the power delivered by the source.
d. Determine the power absorbed by each parallel resistor.
e. Verify Eq. (9).

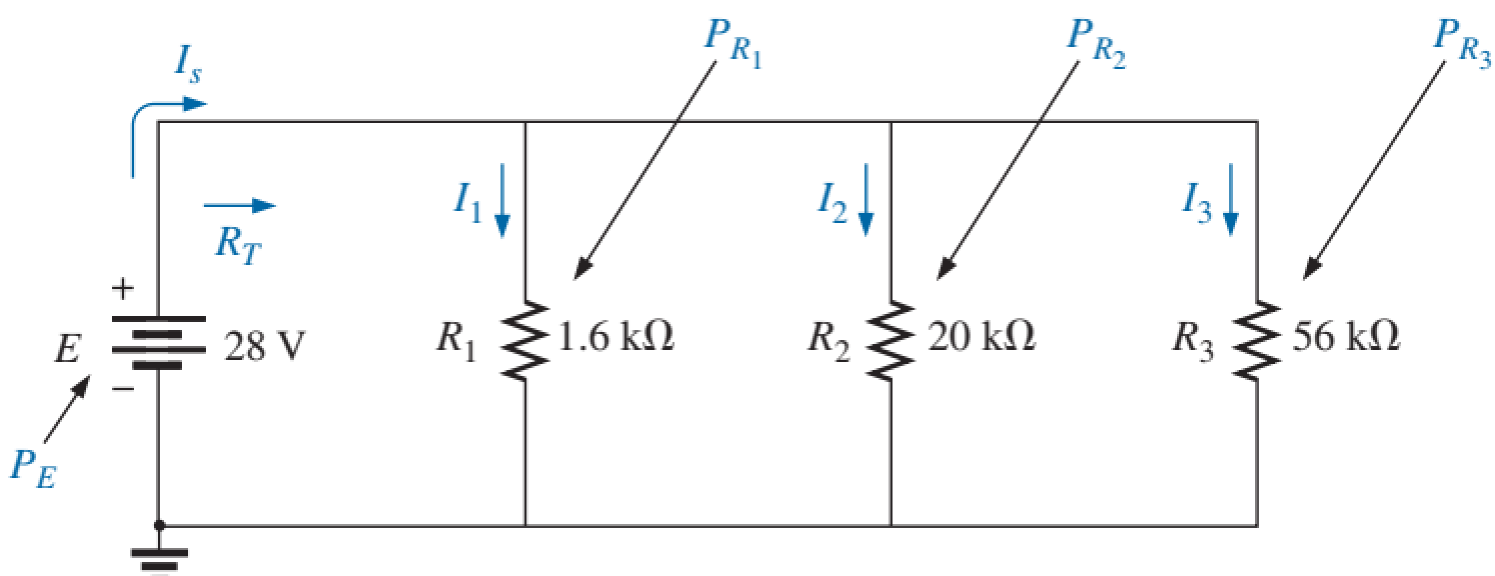

Figure 20

4. Power Distribution in a Parallel Circuit (4/9)

Solutions:

a. $$R_T = \frac{1}{\dfrac{1}{R_1} + \dfrac{1}{R_2} + \dfrac{1}{R_3}} = \frac{1}{\dfrac{1}{1.6\ \text{k}\Omega} + \dfrac{1}{20\ \text{k}\Omega} + \dfrac{1}{56\ \text{k}\Omega}}$$

$$= \frac{1}{625 \times 10^{-6} + 50 \times 10^{-6} + 17.867 \times 10^{-6}} = \frac{1}{692.867 \times 10^{-6}}$$

and $R_T = \mathbf{1.44\ k\Omega}$

b. Applying Ohm's law gives

$$I_s = \frac{E}{R_T} = \frac{28\ \text{V}}{1.44\ \text{k}\Omega} = \mathbf{19.44\ mA}$$

4. Power Distribution in a Parallel Circuit (5/9)

Applying Ohm's law again gives

$$I_1 = \frac{V_1}{R_1} = \frac{E}{R_1} = \frac{28\text{ V}}{1.6\text{ k}\Omega} = \mathbf{17.5\text{ mA}}$$

$$I_2 = \frac{V_2}{R_2} = \frac{E}{R_2} = \frac{28\text{ V}}{20\text{ k}\Omega} = \mathbf{1.4\text{ mA}}$$

$$I_3 = \frac{V_3}{R_3} = \frac{E}{R_3} = \frac{28\text{ V}}{56\text{ k}\Omega} = \mathbf{0.5\text{ mA}}$$

c. Applying Eq.(10). gives

$$P_E = EI_s = (28\text{ V})(19.4\text{ mA}) = \mathbf{543.2\text{ mW}}$$

d. Applying each form of the power equation gives

$$P_1 = V_1I_1 = EI_1 = (28\text{ V})(17.5\text{ mA}) = \mathbf{490\text{ mW}}$$

$$P_2 = I_2^2R_2 = (1.4\text{ mA})^2(20\text{ k}\Omega) = \mathbf{39.2\text{ mW}}$$

$$P_3 = \frac{V_3^2}{R_3} = \frac{E^2}{R_3} = \frac{(28\text{ V})^2}{56\text{ k}\Omega} = \mathbf{14\text{ mW}}$$

A review of the results clearly substantiates the fact that the larger the resistor, the less is the power absorbed.

e. $$P_E = P_{R_1} + P_{R_2} + P_{R_3}$$

$$\mathbf{543.2\text{ mW}} = 490\text{ mW} + 39.2\text{ mW} + 14\text{ mW} = \mathbf{543.2\text{ mW}} \quad \text{(checks)}$$

4. Power Distribution in a Parallel Circuit (6/9)

Homework!

Example 14: Determine currents I_3 and I_5 in Fig. 21 through application of Kirchhoff's current law (KCL).

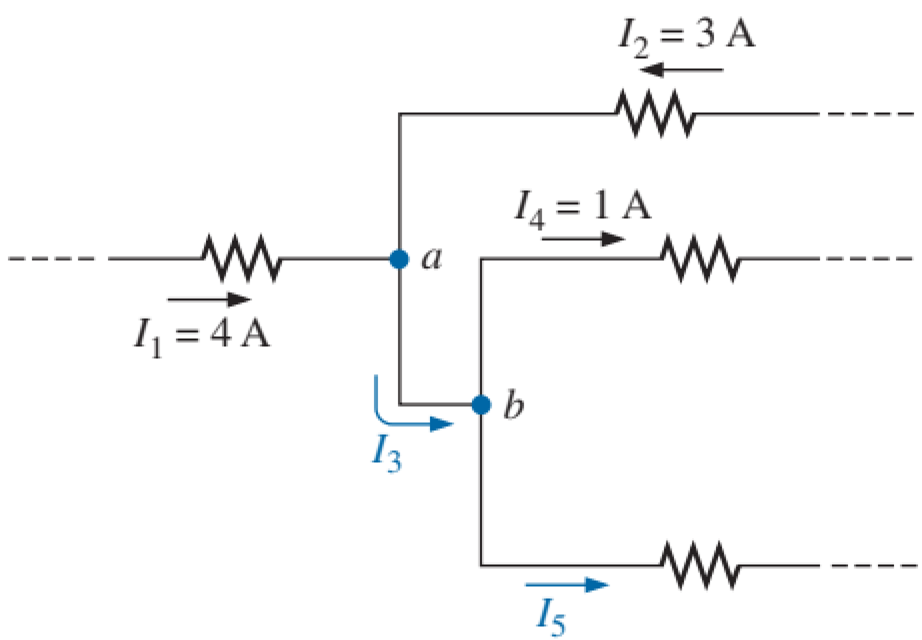

Figure 21

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3. *Charles K.Alexander, Matthew N.O.Sadiku (2017). Fundamentals of Electric Circuits. 6th Edition. McGraw Hill Education.*